KAPLAN

Introduction to Data Communications

Topic 07 – *L07 Ethernet Protocol, Switching, VLANs, and Inter-VLAN Routing*

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Topic 7 – Learning Outcomes

VLAN

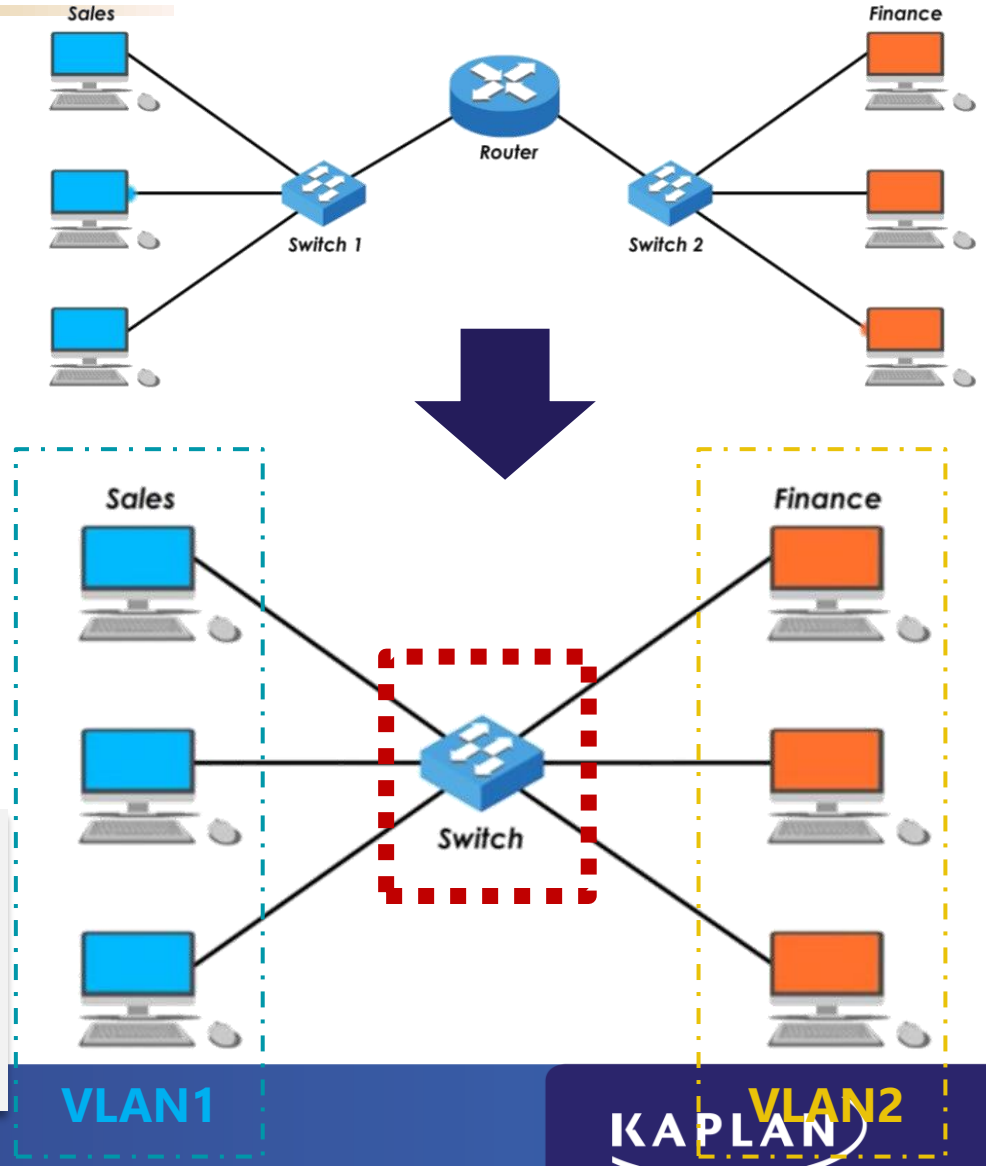
- Fundamentals
- VLAN Types
- IEEE802.11q Standard
  - -Access Ports
  - -VLAN Tag

Virtual LAN (VLAN)

- VLAN is a logical network that group devices/users regardless of their different physical locations
- **VLAN is created at the switch**
- **Why VLAN**
  - Network Segmentation
  - Security
  - Traffic Prioritizing
  - Network Management

Network Design, Deployment, Troubleshooting without affecting other VLANs

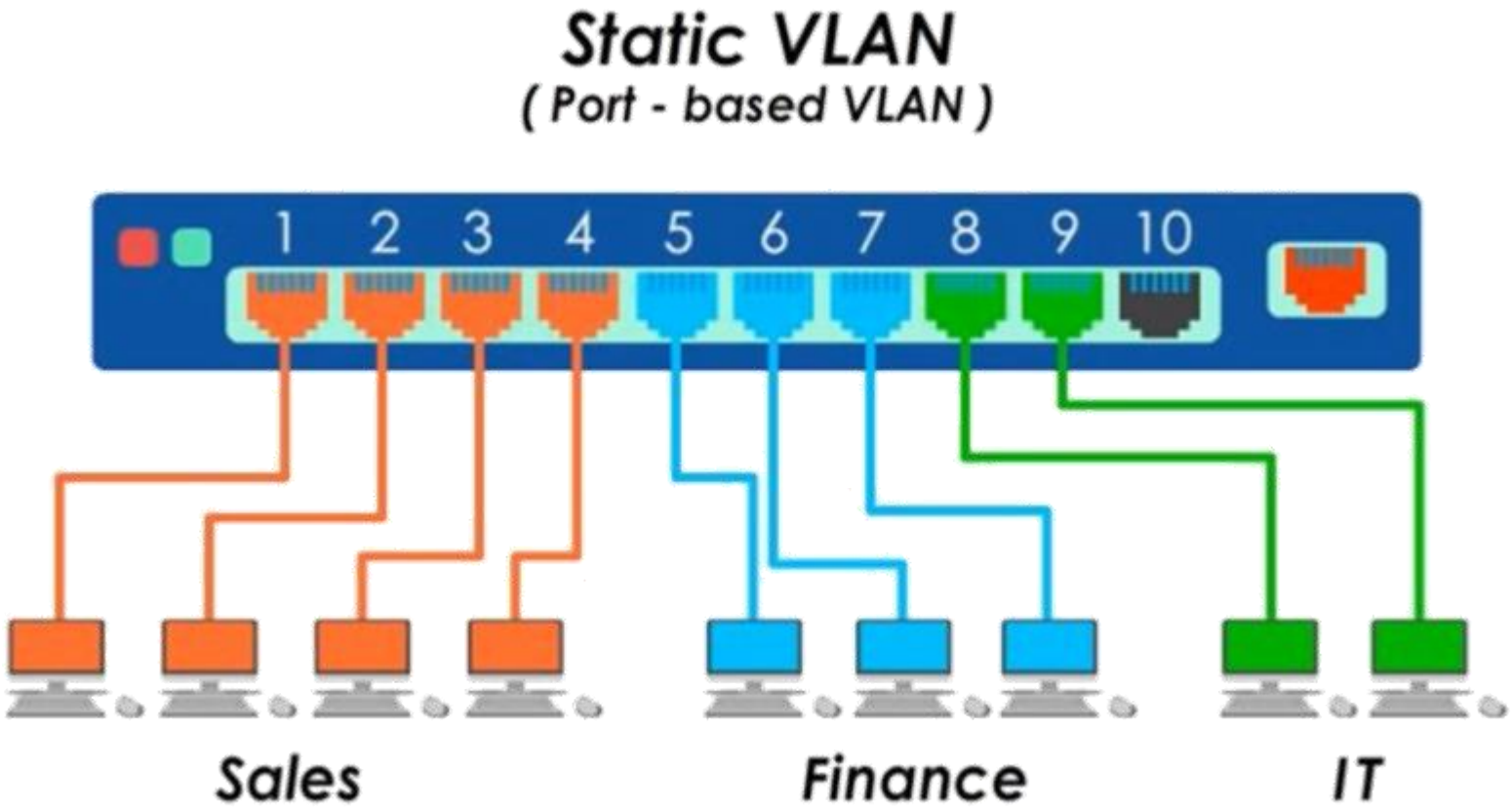
- **VLAN Types:**
  1. Static VLAN
  2. Dynamic VLAN



- inter-vlan-routing | option 1

Type: Static VLAN

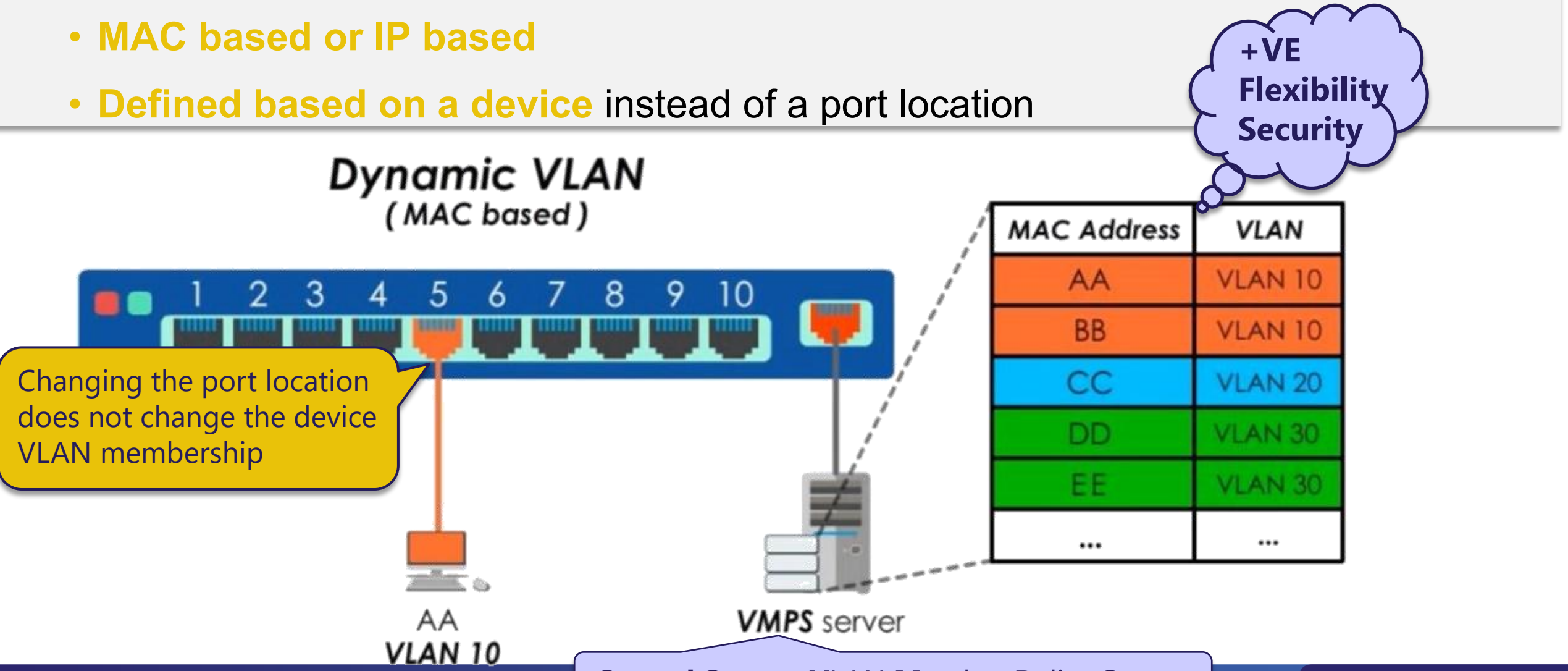
- **Port based**
- **Manual Assignment** of individual ports on a switch to a VLAN



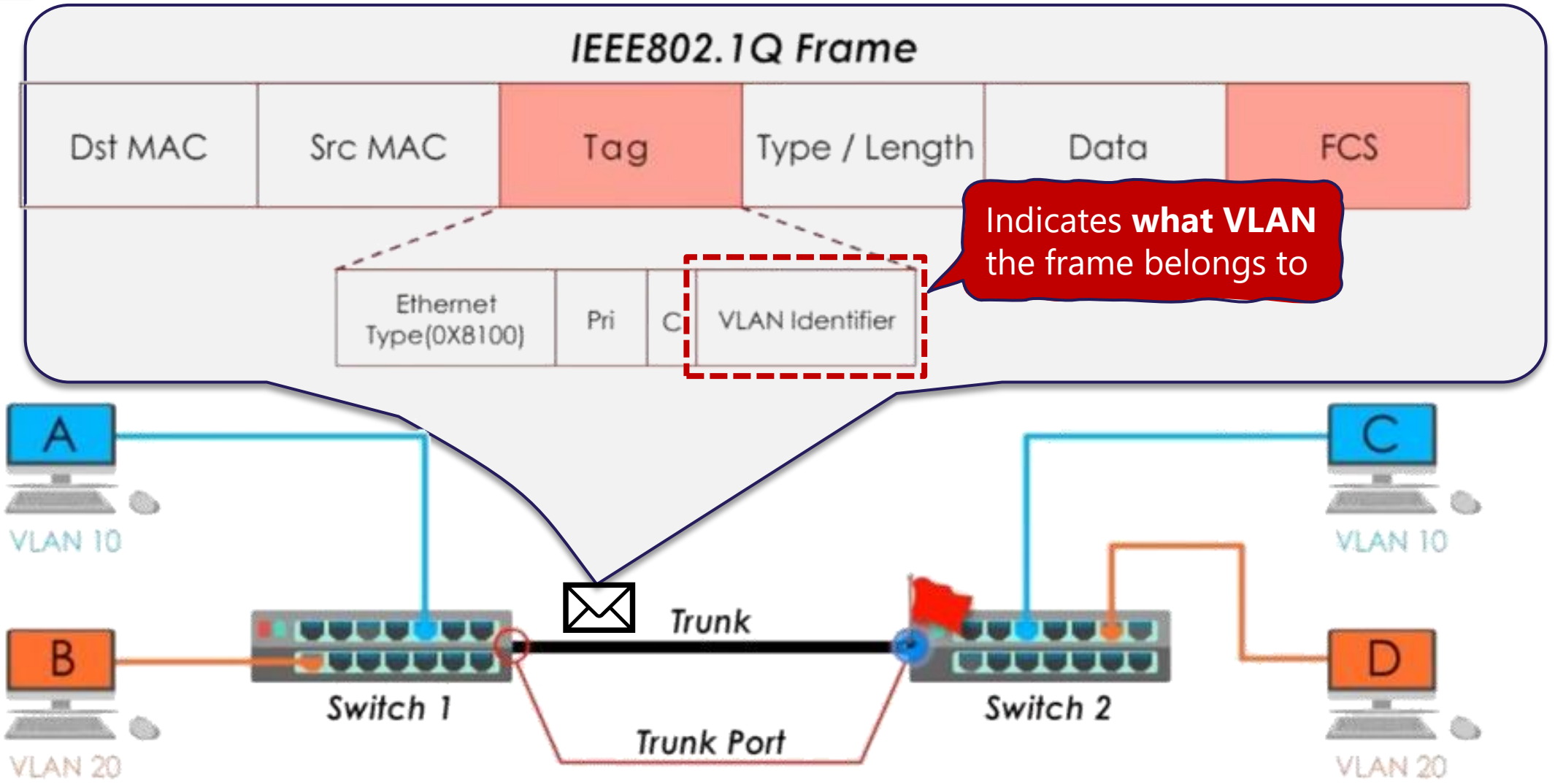
Sales VLAN cannot communicate with Finance or IT VLAN vice versa

Type: Dynamic VLAN

- **MAC based or IP based**
- **Defined based on a device** instead of a port location



VLAN Tag


IEEE802.1Q Frame
Dst MAC
Src MAC
Tag
Type / Length
Data
FCS
Ethernet Type(0X8100)
Pri
C
VLAN Identifier
Indicates **what VLAN** the frame belongs to
A
VLAN 10
B
VLAN 20
Switch 1
Trunk
Trunk Port
Switch 2
C
VLAN 10
D
VLAN 20

Inter-VLAN Routing Fundamentals:

1. Inter-VLAN routing is the process of forwarding traffic between different VLANs within a network.
2. By default, devices within the same VLAN can communicate directly with each other, but communication between devices in different VLANs requires routing.
3. Inter-VLAN routing can be accomplished using routers or layer 3 switches.

Inter-VLAN Implementation Option 1:

- Dedicated Router: In this option, a router with multiple interfaces (one per VLAN) is used to route traffic between VLANs.
- Each VLAN is assigned a separate subnet, and the router serves as the default gateway for devices in each VLAN.
- The router examines the destination IP address of incoming packets and forwards them to the appropriate VLAN interface.

Option 1: Dedicated Router (Traditional Inter-VLAN Routing)

- Each VLAN is connected to a separate physical router interface.
- The router performs routing between VLANs using different physical interfaces.
- Requires multiple router interfaces, which can be inefficient.

```
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown

Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
```

(Repeat for each VLAN)

Example Setup:
VLANs Configuration on the Switch

```
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)# exit

Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown

Switch(config)# interface GigabitEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
```

Router Configuration

```
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown

Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
```

Inter-VLAN Implementation Option 2 (Router-on-a-Stick):

1. Router-on-a-Stick (ROAS): In this option, a single router interface is configured to trunk multiple VLANs over a single physical connection to the switch.
2. The router interface is configured with subinterfaces, each representing a VLAN, and each subinterface is assigned an IP address within the respective VLAN subnet.
3. The switch port connected to the router is configured as a trunk port, allowing it to carry traffic for multiple VLANs tagged with VLAN IDs.

Option 2: Router-on-a-Stick (ROAS)

- Uses a single physical interface on the router with subinterfaces for each VLAN.
- The interface must be configured as a trunk to carry multiple VLANs.

```
Router(config)# interface GigabitEthernet0/0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface GigabitEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config)# interface GigabitEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

(Repeat for additional VLANs)

Example Setup: Switch Configuration

```
Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# no shutdown
```

Router Configuration

```
Router(config)# interface GigabitEthernet0/0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface GigabitEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

Router(config)# interface GigabitEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

Inter-VLAN Implementation Option 3:

- Layer 3 Switch: In this option, a layer 3 switch is used to perform inter-VLAN routing.
- The layer 3 switch has the capability to route traffic between VLANs directly at wire speed without needing an external router.
- Each VLAN is assigned an SVI (Switched Virtual Interface), which acts as the default gateway for devices in the VLAN.
- The layer 3 switch uses routing tables to forward traffic between VLANs based on destination IP addresses.

Option 3: Layer 3 Switch (Inter-VLAN Routing on Switch)

- A Layer 3 switch routes traffic between VLANs using SVIs (Switched Virtual Interfaces).
- Eliminates the need for an external router.

```
Switch(config)# ip routing

Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown

Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
```

(Repeat for additional VLANs)

Example Setup: Enable Routing on the Switch

```
Switch(config)# ip routing
```

Create VLANs & Assign IPs

```
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
```

Assign VLANs to Interfaces

```
Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown

Switch(config)# interface GigabitEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
```

Configure SVIs for Routing

```
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown

Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
```

Multi-Layer Switch vs Router

- **Multi-Layer Switch (MLS)**:
  - A multi-layer switch operates at both layer 2 (data link layer) and layer 3 (network layer) of the OSI model.
  - It can perform inter-VLAN routing and switching functions within a single device, offering high performance and scalability.
  - MLS typically uses specialized hardware and ASICs (Application-Specific Integrated Circuits) for efficient packet forwarding.

Router:

- A router operates at layer 3 (network layer) of the OSI model and is primarily used for routing packets between different networks.
- Routers can perform inter-VLAN routing but may have lower performance compared to multi-layer switches, especially for high-speed LAN environments.
- Routers offer more flexibility and advanced routing capabilities, making them suitable for connecting multiple LANs or WANs.

Thank you

